

Machine Learning models evaluation on Alzheimer Dataset*

Abstract—This report details the exploration and evaluation of various machine learning models, including EfficientB0, EfficientB3, ResNet101, ResNet18, and a Convolutional Neural Network (CNN) built from scratch. The project aimed to address [insert specific problem or task]. Through extensive experimentation and evaluation, the performance of each model was assessed using various evaluation metrics. Challenges encountered during the project are discussed, alongside insights gained from the results.

Index Terms—Alzheimer’s Disease, Deep Learning, MRI, DenseNet169, EfficientNetB1, EfficientNetB4, Fine-Tuning.

I. INTRODUCTION

The rise of machine learning and its applications in diverse domains has spurred interest in developing robust models for [insert problem or task]. This paper investigates the efficacy of different machine learning models in tackling this problem. We present an overview of the problem, motivation behind the research, contributions of the study, and a roadmap of the paper. Alzheimer’s Disease (AD) is a progressive neurodegenerative disorder that impairs memory and cognitive function. Early and accurate diagnosis is crucial for effective management and treatment. Magnetic Resonance Imaging (MRI) is a non-invasive imaging technique widely used in AD diagnosis. Leveraging deep learning for automated classification of MRI images holds promise for enhancing diagnostic accuracy and efficiency. This paper explores the performance of three state-of-the-art pre-trained convolutional neural networks (CNNs) fine-tuned for the classification of Alzheimer’s stages. The contributions of this paper include: (1) a detailed comparison of DenseNet169, EfficientNetB1, and EfficientNetB4 models on a preprocessed MRI dataset, (2) insights into the challenges faced during the fine-tuning process, and (3) a comprehensive analysis of the results. The remainder of this paper is organized as follows: Section II reviews related work, Section III describes the dataset and preprocessing steps, Section IV details the methodology, Section V presents the experimental results, and Section VI concludes the study.

II. RELATED WORK

Several studies have applied deep learning techniques to classify Alzheimer’s Disease using MRI images. For instance, Sarraf et al. (2016) utilized a pre-trained AlexNet model to classify AD from functional MRI data, achieving significant accuracy improvements. Similarly, Ding et al. (2019) explored the use of VGG and ResNet architectures for AD classification, demonstrating the effectiveness of transfer learning in medical image analysis. This study builds upon these works

by evaluating newer architectures, specifically DenseNet169, EfficientNetB1, and EfficientNetB4, and fine-tuning them on a specific AD dataset.

]Data Description and Preprocessing Steps

A. Model Architecture

Three pre-trained models—DenseNet169, EfficientNetB1, and EfficientNetB4—were fine-tuned for this classification task. Each model was modified by adding a global max pooling layer, a flatten layer, and two dense layers (256 units with ReLU activation, followed by a softmax layer for classification).

B. Fine-Tuning

To fine-tune the models, the base layers of each pre-trained network were initially frozen to retain the learned features from ImageNet. Subsequently, the last 10 layers were unfrozen and retrained on the Alzheimer’s dataset with a reduced learning rate of $1e-5$.

Data Augmentation Data augmentation was applied using the ImageDataGenerator class to generate batches of tensor image data with real-time data augmentation. This included random rotations, shifts, shear, zoom, and flips to improve the model’s generalization ability.

Training Models were trained using the Adam optimizer with categorical cross-entropy loss. Each model was trained for 10 epochs with a batch size of 32, utilizing a validation split of 20

C. Methodology

Our approach to addressing the problem involved training and evaluating multiple machine learning models, including EfficientB0, EfficientB3, ResNet101, ResNet18, and a custom CNN architecture. The data was split into training, validation, and testing sets, with hyperparameters tuned using [insert method]. A graphical user interface (GUI) was developed to facilitate model deployment and interaction.

D. Results

The experimental results demonstrate varying performances among the different models. Evaluation metrics such as accuracy, precision, recall, and F1-score were used to assess model performance. The results are presented and discussed in detail, highlighting the strengths and weaknesses of each model. Figures, tables, and diagrams are utilized to visualize the findings.

E. Conclusion

In conclusion, this study provides insights into the effectiveness of different machine learning models for Classifying Alzheimer diseases. Future research could focus on increasing the accuracy of the models. Overall, this study contributes to the growing body of literature on classifying diseases using machine learning and underscores the importance of selecting appropriate models for specific tasks.

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